

Juan David Muñoz Bolaños

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

jmunozbolanos@gmail.com

May 25, 2026

IMS Nanofabrication GmbH

Recruiting Team

Dresdner Str. 47, 1200 Vienna, Austria

Application for the position: Development Engineer

Dear Recruiting Team,

IMS Nanofabrication drives the limits of lithography at the nanometer scale. As an optical systems engineer finishing my PhD at the Medical University of Innsbruck, I have spent the last four years building complex optomechatronic systems (advanced microscopes) and developing data analysis pipelines to optimize their performance. My background in physics engineering, combined with my exposure to industrial semiconductor workflows at the ASML and ZEISS summer schools, makes me eager to contribute to the development of your Multi Beam Mask Writer systems.

In my current research, I designed and built an automated wavefront sensing and adaptive optics system from scratch. This involved integrating high precision optical components, spatial light modulators, and scientific cameras, while writing the entire control and data acquisition stack in Python and C++. To ensure the system achieved the best potential performance, I developed iterative phase retrieval and optimization algorithms to process wavefront data in real time. This hands-on experience in combining hardware control with advanced signal processing directly aligns with your need for validating new features and analyzing complex log and metrology data.

Beyond system development, I focus on structured testing and quality frameworks. I have structured my instrument development following V-Model systems engineering principles, ensuring clear requirements tracking from initial concept to final validation. By applying Design of Experiments and systematic optical tolerance analysis in python, I have successfully bridged the gap between early stage R&D and stable, reproducible system performance.

I will complete my PhD in June 2026. I am Colombian originally and hold an unrestricted working resident permit in Austria (rot-weiss-rot-plus). I look forward to discussing how my background in optomechanical and electronic systems, data analysis, and structured testing can support your team in Vienna.

Best regards,

Juan David Munoz Bolanos

Juan David Munoz Bolanos

Junior Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

munozbolanos@gmail.com



Summary

Junior system engineer with hands-on lab work building, aligning, and calibrating optical laser systems using spatial light modulators and holography. I'm comfortable with PC and control technology, used to digging into technical problems systematically.

Research/Professional Experience

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Optical Systems Engineer / PhD Researcher

Optomechatronic System Design: Designed and built a multiphoton microscope and an automated wavefront sensing system from scratch, integrating lasers, spatial light modulators, and deformable mirrors.

Data Analysis & Algorithms: Developed Python and C++ pipelines for real time signal processing, iterative phase retrieval, and wavefield simulations to optimize system performance.

Systems Engineering & Quality: Applied V Model methodologies, Design of Experiments, and FMEA to track requirements and systematically resolve technical defects during prototype iterations.

Lab Management & Safety: Managed a high power laser laboratory, ensuring EH&S compliance, maintaining equipment logs, and coordinating component sourcing with suppliers like Thorlabs and Edmund Optics.

Aug 2018 to Oct 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision Engineer (Field Service)

Machine Vision Pipelines: Engineered and deployed Halcon and Cognex VisionPro image processing pipelines for high speed industrial production lines, improving quality control.

Illumination Design: Designed custom bright field and dark field illumination systems to reliably detect defects, liquid levels, and sealing states on fast moving products.

Industrial Automation: Integrated vision systems with robotic arms and PLC controllers, executing on site testing, validation, and customer sign off.

Technical Competencies

Software & Tools

PythonC++LabVIEWHalconOpenCVZEMAXMATLABGit

Optical Systems

Adaptive OpticsWavefront SensingMicroscopySpectroscopyLaser SystemsDMDs

Systems & Quality

V-Model8D Problem SolvingDesign of ExperimentsISO 9001

Education

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

Oct 2019 to Sep 2021	Friedrich Schiller Univ. Jena Master of Science in Photonics	Jena, Germany
Aug 2012 to Aug 2018	University of Cauca Bachelor of Science in Physics Engineering	Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

- Biking/Hiking
- Bachata Dancing
- Harmonica
- Learning German

Languages

- English (Fluent)
- German (Intermediate B2)
- Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte Medical University of Innsbruck monika.ritsch-marte@i-med.ac.at	PD Dr. Christoph Krafft Leibniz IPHT christoph.krafft@leibniz-ipht.de
--	--

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR e INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

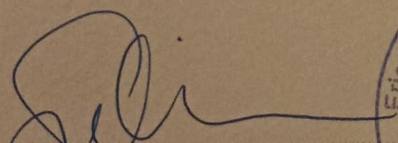
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

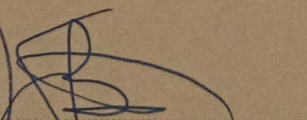
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

